

Two Incomparable Denominator Exclusions for $\sum_{n \geq 2} (n! - 1)^{-1}$

A divisibility bound, a size bound, an adjacent-unit no-go, and exact rationality coordinates for Erdős Problem #68

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ABSTRACT

Erdős asked whether

$$S = \sum_{n \geq 2} \frac{1}{n! - 1}$$

is irrational. This problem remains open. What is proved here are two exclusions on any displayed rational representation $S = a/q$ with $q > 0$, and they are incomparable. The first is a divisibility exclusion,

$$q \nmid 299999!,$$

so some prime power in q exceeds its multiplicity in 299999!. The second is a size exclusion,

$$q \geq 2^{39990} > 10^{12038},$$

Together these restrict the denominator's size and prime-power multiplicities. Neither restriction implies the other. The first rests on a Lean-checked consumer together with one exact GMP carry computation, 300000 $\nmid Z_{300000}$; the second on 23449 certified continued-fraction partial quotients from an 80000-bit exact rational bracket. Both computations use exact integer arithmetic.

A third unconditional result is a no-go. Two consecutive unit carries reduce, after an exact denominator telescope, to the universal inequality $0 < U_{m+2} \leq V_{m+2}$. A divisibility obstruction derived only after assuming those two carries therefore cannot rule them out, and the adjacent-unit cancellation route is closed.

The exclusions are read off exact coordinates for the problem, which we state next because they fix the missing quantifier rather than because they advance it. Put $H_m = \sum_{2 \leq n \leq m} (n! - 1)^{-1}$ and $Z_m = \lfloor m!H_m \rfloor + 1$, and let b_m be the exact integer carry in the recurrence $Z_m = mZ_{m-1} + 1 - b_m$. Lean proves that S is rational precisely when $b_m = 1$ for all sufficiently large m , and $b_m = 1$ is equivalent to $m \mid Z_m$ for every $m \geq 3$, so

$$S \notin \mathbb{Q} \iff (\forall B)(\exists m > B) m \nmid Z_m.$$

A companion series gives a second coordinate. With $C = \sum_{n \geq 2} (n!(n! - 1))^{-1}$ the termwise identity $C + (e - 2) = S$ yields

$$S \in \mathbb{Q} \iff \lfloor m!C \rfloor \equiv -2 \pmod{m} \text{ for all sufficiently large } m,$$

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so irrationality of S is cofinal escape of one explicit factorial orbit from one distinguished residue class. Neither equivalence proves that the escape occurs.

A further reduction describes cancellation at a fixed prime. If q^e is the largest power of a prime q occurring among the displayed denominators $n! - 1$, then the common-denominator numerator modulo q is a q -adic unit times the sum of the reciprocal maximal-hit cofactors. Hence the complete power q^e survives prefix reduction exactly when this principal residue is nonzero. Complete cancellation occurs at the Wilson prefixes for $q = 139$ and $q = 2593$; the corresponding residue identities are also checked in Lean. The set of maximal hits therefore does not determine whether q^e survives reduction: their cofactor residues matter.

The remaining problem is to prove $m \nmid Z_m$ for arbitrarily large m . None of the finite computations, exclusions, or structural reductions in this paper proves that assertion.

1 The problem

Problem 1.1 (Erdős #68). Is

$$S = \sum_{n \geq 2} \frac{1}{n! - 1}$$

irrational?

Main results at a glance

Result	Exact scope
Fixed companion-orbit boundary	For $C = \sum_{n \geq 2} (n!(n! - 1))^{-1}$, one has $S \in \mathbb{Q}$ iff $[m!C] \equiv -2 \pmod{m}$ eventually; equivalently, irrationality is cofinal escape from that residue.
Exact irrationality criterion	$S \notin \mathbb{Q}$ iff for every B some $m > B$ satisfies $m \nmid Z_m$.
Finite denominator exclusions	Two incomparable statements: $q \nmid 299999!$ from the carry certificate and $q \geq 2^{39990} > 10^{12038}$ from certified continued fractions.
Prime-pole cancellation law	For a finite prefix and prime q with positive attained maximal exponent e , the complete power q^e survives reduction iff the reciprocal sum of its maximal-hit cofactors is nonzero. Exact scans give full cancellations at $q = 139$ and 2593 ; Lean checks the residue equalities.
Adjacent-unit obstruction	After the pair assumption, the complete width-one window cancels to the universal reduced-numerator bound; it cannot exclude the pair.
Moving-factor criterion	One moving prefix-private prime plus two explicit scale bounds implies irrationality; arbitrary split factors give a stronger normalized-collision criterion, while every fixed owner pair is eventually absorbed. The cofinal scale and disagreement producers remain open.
Further reductions	Checked channel identities and projection lemmas state the additional estimates that those methods would require.
Open step	No cofinal miss or cofinal strict residual nonvanishing is proved; the irrationality question remains open.

Erdős states the problem on p. 102 of his 1988 survey and, in the same passage, records the broader expectation that $\sum_n 1/(n! + t)$ is irrational, indeed transcendental, for every integer t [1, p. 102]. Erdős presents this broader statement as a conjecture.

Numbering and current status follow [Bloom's Erdős problem catalogue](#) [2]. The problem is open. The companion series $\sum_{n \geq 0} 1/n! = e$ and $\sum_{n \geq 2} 1/(n! + 1)$ sit in the same family, and the difficulty here is the same one that makes the Erdős–Borwein constant hard: the denominators $n! - 1$ grow fast enough that convergence is trivial and slow enough, in the arithmetic sense, that no single congruence controls them.

This note is one of the problem-owned series. It repeats the definitions and the claim boundary it needs, so that a reader who arrives at #68 alone does not have to assemble them from the joint integration monograph.

Status. The problem treated here is open, and this note does not close it. Every statement below marked as checked is a proposition that the pinned Lean kernel accepts from the sources this note links to, with no sorry, no added axiom, and no unchecked evaluation. That is a claim about the formal statement, not about its mathematical interest, its novelty, or the original problem. The unresolved obligations are named exactly,

in their own section, and none of the finite computations, reductions, or no-go results here removes one of them.

Companion system context. The [claim and trust boundary](#), [cold-clone route to proof authority](#), and [public contribution protocol](#) are described in sibling papers. Those descriptions do not change the mathematical status of this note.

Statement	Status	Exact boundary
Irrationality of S	Open	No proof is claimed.
Fixed companion-orbit boundary	Checked	The two endpoint equivalences are checked in <code>CompanionOrbitRationality.lean</code> ; their exact declaration names and source routes appear in Appendix A. Both depend exactly on <code>propext</code> , <code>Classical.choice</code> , and <code>Quot.sound</code> ; neither proves that the cofinal misses occur.
Canonical factorial digit kernel	Checked	Floor formula, digit bounds, remainder recurrence, finite expansion, zero-tail propagation.
Channel integrality	Checked	$(d!)^{\lfloor i/d \rfloor} \mid i!$, with exact denominator cancellation.
Channel congruence and LCM tax	Checked	$V_d(\lambda) \equiv M \pmod{d! - 1}$; annihilating channels through D forces $L_D \mid M$.
Prime unit translator	Checked	The pair $(p, -1)$ on $(p - 1, p)$ has moment 0, all channels 0 except $d = p$, and p -channel numerator $p! - 1$.
Weighted projection rigidity	Checked	If $Z \equiv T \pmod{R}$, $Q_i \mid R$, and $Z \leq B < Q_i$, then unequal residues $T \bmod Q_1$ and $T \bmod Q_2$ exclude that endpoint.
Factor-split projection reduction	Checked	Two divisor factors of one private modulus support the same cancellation and disagreement bounds; coprime factors give a branch-free floor and may lie in one private quotient.
Zero plateau and first-exit carry	Checked	Grid threshold, plateau equality of grid integers, forced zero digit, carry $b \in \{0, -1\}$.

Statement	Status	Exact boundary
Prime-power prefix obstruction	Checked	Rationality forces p^k to divide the strict successor at kp whenever the factorial clears the denominator and p is coprime to it.
Exact carry characterization	Checked	The normalized strict successors converge to S ; S is rational exactly when $b_m = 1$ eventually, equivalently S is irrational exactly when non-unit carries occur cofinally.
Explicit denominator exclusions	Checked implications; exact finite certificates	Exact reduction gives $60 \nmid Z_{60}$, $64 \nmid Z_{64}$, and $67 \nmid Z_{67}$. The GMP carry computation gives $q \nmid 299999!$ (hence $q \geq 300000$); an independent bracket-certified continued-fraction computation gives $q \geq 2^{39990} > 10^{12038}$. Neither changes a quantifier.
Adjacent-unit cancellation normalizer	Checked	Under consecutive unit carries, the exact offset and denominator share the same positive two-step normalizer, so their window inequality is equivalent to the universal bound $0 < U_{m+2} \leq V_{m+2}$.
Digits eventually zero $\iff S$ rational	Returned derivation	Complete on the return; not yet kernel-checked here.
Factorial-gap lcm growth $\gg N^{4/3} \log N$	Derived, source-verified	Derived below from a cited factorial-congruence theorem; not kernel-checked and not used as an input to any claim below.
Finite certificates ($D = 3, D = 9, D \leq 12$)	Verified finite instances	Each excludes only the denominators it names.
Unbounded strict nonvanishing	Open	Required to turn the channel rounding argument into an irrationality proof.

2 The fixed companion orbit

The exact reduction of this section is a transport of a classical criterion, and the classical criterion should be named first. A real number is rational exactly when its canonical factorial-base digits vanish from some index on; this is due to Cantor [7], and the standard modern account is Galambos [8, Ch. 1]. One direction is the observation that $S = a/b$ makes $m!S$ an integer for $m \geq b$, and the other is a tail bound. What is added below is the transport of that criterion along an explicit companion telescope, the identification of the resulting digit value, and the observation that the same shape survives for the whole family $\sum_n (n! + t)^{-1}$ after a sign adjustment. The difficulty is

entirely in the unproved cofinal-miss producer, so this is a reduction and not an advance on the endpoint.

The reduction uses one fixed real number rather than the changing rational prefixes. Define

$$C = \sum_{n \geq 2} \frac{1}{n!(n! - 1)}.$$

The elementary identity

$$\frac{1}{n! - 1} = \frac{1}{n!} + \frac{1}{n!(n! - 1)}$$

and absolute convergence give

$$C + (e - 2) = S. \quad (1)$$

Theorem 2.1 (fixed companion-orbit rationality boundary). *The following statements are equivalent:*

1. $S \in \mathbb{Q}$;
2. $(\lfloor m!C \rfloor + 2) \bmod m = 0$ for every sufficiently large m .

Consequently,

$$S \notin \mathbb{Q} \iff (\forall B)(\exists m > B) (\lfloor m!C \rfloor + 2) \bmod m \neq 0. \quad (2)$$

Proof. First suppose $S = a/q$ with $q > 0$, and take m so large that $q \mid (m - 1)!$. Write

$$E_m = m! \sum_{2 \leq n \leq m} \frac{1}{n!}, \quad \varepsilon_m = m! \sum_{n > m} \frac{1}{n!}.$$

Then E_m is an integer, $0 < \varepsilon_m < 1$, and $m!S$ is an integer divisible by m . Multiplying (1) by $m!$ therefore gives

$$\lfloor m!C \rfloor = m!S - E_m - 1.$$

Every summand of E_m except the endpoint $m!/m! = 1$ is divisible by m . Thus $E_m \equiv 1 \pmod{m}$ and $\lfloor m!C \rfloor \equiv -2 \pmod{m}$.

Conversely, assume the displayed congruence from some index onward. Let $d_m(C) = \lfloor m!C \rfloor - m \lfloor (m - 1)!C \rfloor$ be the canonical factorial digit. Since $0 \leq d_m(C) < m$, for $m \geq 3$ the congruence is equivalent to

$$d_m(C) = m - 2.$$

Choose N beyond the exceptional indices. The canonical factorial expansion of C then has the form

$$C = \lfloor C \rfloor + \sum_{m=2}^N \frac{d_m(C)}{m!} + \sum_{m > N} \frac{m-2}{m!}.$$

Adding $e - 2 = \sum_{m \geq 2} 1/m!$ and using the telescoping identity

$$\sum_{m > N} \frac{m-1}{m!} = \sum_{m > N} \left(\frac{1}{(m-1)!} - \frac{1}{m!} \right) = \frac{1}{N!}$$

gives

$$S = \lfloor C \rfloor + \sum_{m=2}^N \frac{d_m(C) + 1}{m!} + \frac{1}{N!},$$

which is rational. Negating the eventual statement yields the cofinal formulation above. $\square$

Remark (the exact remaining boundary). The theorem isolates the problem without solving it. A proof of Problem #68 must still show that the orbit $\lfloor m!C \rfloor \bmod m$ does not eventually concentrate at -2 . Finite computation can measure this concentration but cannot discharge the cofinal quantifier.

3 Canonical factorial digits

Write $\theta_0 = \{x\} = x - \lfloor x \rfloor$ and, for $m \geq 1$,

$$d_m = \lfloor m \theta_{m-1} \rfloor, \quad \theta_m = m \theta_{m-1} - d_m.$$

This is the factorial base taken in its canonical form. The kernel checks the floor formula, the digit bounds $0 \leq d_m < m$, the recurrence $\theta_{m+1} = (m+1)\theta_m - d_{m+1}$, the finite telescoping expansion

$$x = \lfloor x \rfloor + \sum_{m=2}^N \frac{d_m}{m!} + \frac{\theta_N}{N!},$$

and the propagation rule: a zero remainder at one index forces every later digit to vanish. These are *canonical digit eq floor mul remainder*, *canonical digit nonneg*, *canonical digit lt radix*, *canonical remainder recurrence*, *factorial expansion partial*, and *zero remainder forces zero digit tail*; they hold for every real x , not only for S .

The returned derivation adds the criterion that makes the representation useful: the digits $d_m(S)$ are eventually zero if and only if S is rational, and this is equivalent to eventual integrality of the factorial tail state. That equivalence is complete on the return and is not yet kernel-checked here. We state it as an equivalence because that is what the derivation gives; we do not use it below as though it were checked.

Note what the criterion is not. Canonical normalisation is an exact reformulation of rationality. It does not by itself supply an obstruction, and a zero digit is not the same event as a zero-branch hit: the returned data contain canonical zero digits at $m = 5$ and $m = 23$, while the zero-branch list is empty through $m = 100000$.

3.1 The finite second layer and the floor-stability boundary

There is a sharper coordinate for the actual carry sequence. The elementary identity

$$\frac{1}{n! - 1} = \frac{1}{n!} + \frac{1}{n!(n! - 1)}$$

splits the prefix as

$$H_m = E_m + P_m, \quad E_m = \sum_{2 \leq n \leq m} \frac{1}{n!}, \quad P_m = \sum_{2 \leq n \leq m} \frac{1}{n!(n! - 1)}.$$

For $m \geq 2$, the first part is integral at factorial scale and satisfies the exact radix recurrence $m!E_m = m((m-1)!E_{m-1}) + 1$. Define the strict-successor carry by $Z_m = mZ_{m-1} + 1 - b_m$. If

$$f_m = \lfloor m!P_m \rfloor, \quad d_m^{(2)} = f_m - mf_{m-1},$$

then Lean proves the finite digit law

$$d_m^{(2)} = m - 1 - b_m. \tag{2.1}$$

After removing the factorial-scale integral prefix E_m of $e - 2$, the strict-successor carry is exactly the defect from the maximal second-layer digit.

Put $S_2 = \sum_{n \geq 2} 1/(n!(n! - 1))$. The passage from P_m to S_2 is the delicate point. Formula (2.1) transfers to the canonical digit of S_2 only when the floors at both scales $m!$ and $(m-1)!$ are stable. Positivity of the omitted tail and an upper bound below one do not suffice: a floor changes whenever the finite prefix already lies closer to the next integer than that tail bound. The source therefore makes the two floor-stability hypotheses explicit. It also records the normalization fact

that canonical factorial digits of a rational number eventually vanish. Hypothetical rationality of S does not imply rationality of $S_2 = S - (e - 2)$, since $e - 2$ is irrational. An exact-Fraction computation checks the relevant next-integer gaps through $m = 500$. A cofinal floor-stability theorem is still needed.

A separate exact reformulation runs through a defect automaton. For a rational centre recurrence $F_m = mF_{m-1} + 1 + \varepsilon_m - C_m$, the kernel checks that the integer ceiling defect code equals $\lfloor m\delta_{m-1} - \varepsilon_m \rfloor$ and that $\delta_m = m\delta_{m-1} - \varepsilon_m - q_m$, with the specialisation $\varepsilon_m = 1/(m! - 1)$ written out. What is checked is the algebra of the automaton. Proving that the finite-sum residual centre satisfies the premise is a separate step and is not done.

A nearby floor criterion gives a precise test for a natural approach. Koepf and Schmersau prove that eventual equality between the floors of n times a partial sum and n times its limit forces irrationality [5, Theorem 1.1, p. 117]; their rational-term version obtains that equality from prefix integrality at a scale p_n and the strict tail bound $a - s_n < 1/(np_n)$ [5, Theorems 2.2–2.3, pp. 119–120]. For the natural termwise clearing choice

$$p_n = \text{lcm}\{k! - 1 : 2 \leq k \leq n\},$$

the last two denominators already show the obstruction:

$$p_n \geq \frac{(n! - 1)((n - 1)! - 1)}{n - 1},$$

because $\gcd(n! - 1, (n - 1)! - 1) = \gcd((n - 1)! - 1, n - 1) \leq n - 1$. For $n \geq 4$, the first omitted summand $1/((n + 1)! - 1)$ is then already larger than $1/(np_n)$, so this natural p_n cannot satisfy their tail hypothesis. Cancellation in the reduced prefix denominator could in principle give a smaller scale, but proving enough cancellation is another form of the present denominator problem. This calculation explains exactly why their hypothesis fails for the natural clearing denominator.

Duverney's fast-series criteria fail at a different, equally exact boundary. His Theorem 3.1 assumes two-sided quadratic denominator growth $cu_n^2 \leq u_{n+1} \leq c'u_n^2$, while for $u_n = n! - 1$ one has $u_{n+1}/u_n^2 \rightarrow 0$ [6, pp. 275, 285–286]. The all-positive specialization in Corollary 3.2 additionally requires

$$\sum_n \left| \frac{u_{n+1}}{u_n^2} - 1 \right| < \infty,$$

whereas the summands tend to one here [6, Corollary 3.2, p. 287]. Neither criterion applies.

The sharp recent theorem of Barreto, Kang, Kim, Kovač, and Zhang has a similarly explicit ceiling. Its $d = 1$ case proves irrationality of $\sum_n 1/a_n$ when $a_n^{1/2^n} \rightarrow \infty$, whereas

$$(n! - 1)^{1/2^n} \rightarrow 1$$

for the present choice $a_n = n! - 1$ [10, Theorems 2–3, pp. 2–4]. The proof nevertheless identifies a useful exact condition: Mahler's elementary rationality floor is contradicted by prefix-clearing integers D_N for which the cleared positive tails satisfy $\liminf_N D_N r_N = 0$ [10, Lemma 8 and Proposition 12, pp. 6, 9–12]. The ordinary product of the factorial-gap denominators is far too large for that estimate; a transfer would need a low-height clearing subsequence, or enough exact cancellation in their least common multiple. Thus the new theorem supplies a precise target inequality and adaptive-cutoff architecture, but not the missing arithmetic bound.

The ordinary factorial-series direction survives more usefully. Dividing the strict-successor recurrence $Z_m = mZ_{m-1} + 1 - b_m$ by $m!$ and telescoping gives the exact finite identity

$$\frac{Z_M}{M!} = \frac{Z_2}{2!} + \sum_{m=3}^M \frac{1 - b_m}{m!}.$$

Thus the carry defects $1 - b_m$ are genuine factorial-series coefficients. Hančl and Tijdeman give exact rationality classifications for polynomial coefficients and finite-difference criteria for broader ordinary factorial series [9, Theorem 3.1 and Corollary 3.1, pp. 390–391]. Their denominator is the cumulative linear product $\prod_{n \leq N} (an + b)$, not the individual number $N! - 1$. Applied to the display above, the classical Cantor–Oppenheim criterion still needs $1 - b_m \neq 0$ infinitely often. Proving this is exactly the unresolved cofinal non-unit-carry assertion.

4 Integral channels and the LCM tax

Fix $d \geq 2$. The kernel proves the divisibility

$$(d!)^{\lfloor i/d \rfloor} \mid i! \quad (i \geq 0),$$

defines the integral channel weight $W_{d,i}$ obtained by cancelling that factor, and checks the exact cancellation. It also checks the consecutive channel event

$$n W_{d,n-1} - W_{d,n} = 0 \quad \text{whenever } d \nmid n.$$

So the channel weight is arithmetically inert except at multiples of d . The checked source statements are *factorial pow floor dvd factorial*, *channel weight mul denominator*, and *channel event eq zero of not dvd*.

Let λ be a finitely supported integer vector on indices $n \geq 2$, let $M = M(\lambda)$ be its factorial moment, and let $V_d(\lambda)$ be the d -th channel numerator. The kernel checks two facts about them.

Theorem 4.1 (channel congruence). *For every finite integer support and every $d \geq 2$,*

$$V_d(\lambda) \equiv M(\lambda) \pmod{d! - 1}.$$

Consequently $d! - 1$ divides $M(\lambda) - V_d(\lambda)$, a vanishing d -th channel forces $(d! - 1) \mid M(\lambda)$, and annihilating every channel $2 \leq d \leq D$ forces

$$L_D = \text{lcm}_{2 \leq d \leq D} (d! - 1) \mid M(\lambda).$$

Theorem 4.2 (integral normal form). *For every finite integer support and every $d \geq 2$ there is an integer k with $V_d(\lambda) = M(\lambda) + (d! - 1)k$.*

Theorem 4.2 is the sharper of the two for design purposes. It says that every zero-moment variation of the support changes the normalised d -th channel contribution by an integer only. The support-matroid layer therefore cannot manufacture an extra fractional cancellation coordinate: the congruence forces every normalised channel defect to be integral.

Theorem 4.1 imposes an arithmetic cost. Any localiser that kills the low channels in order to isolate the tail has moment divisible by L_D , and L_D grows faster than the tail shrinks. The relevant quantitative estimate is

$$\log \text{lcm}_{2 \leq n \leq N} (n! - 1) \gg N^{4/3} \log N \quad (N \rightarrow \infty).$$

Here is the source-checked derivation. Put

$$Q_N = \prod_{2 \leq n \leq N} (n! - 1), \quad L_N = \text{lcm}_{2 \leq n \leq N} (n! - 1).$$

For an odd prime p , let m_p count the indices $2 \leq n \leq N$ for which $p \mid n! - 1$. Such an index necessarily satisfies $n < p$. The factorial congruence multiplicity estimate of Garaev, Luca, and

Shparlinski [3, arXiv v1, Thm. 12, p. 16], applied on the interval $1 \leq n \leq \min(N, p-1)$, therefore gives $m_p \ll N^{2/3}$. (The prime 2 divides none of these factors.) If

$$E_p = \max_{2 \leq n \leq N} v_p(n! - 1),$$

then

$$\log Q_N = \sum_p \sum_{n=2}^N v_p(n! - 1) \log p \leq (\max_p m_p) \sum_p E_p \log p \ll N^{2/3} \log L_N.$$

On the other hand, Stirling summation gives $\log Q_N \asymp N^2 \log N$, proving the displayed lower bound for $\log L_N$. The cited source states the multiplicity theorem. The lcm corollary is derived here and has not been formalized in Lean. It is recorded because it quantifies the denominator growth encountered by this method. No later argument depends on it.

The primitive lcm divisibility after cofactor removal is also checked: factorial valuations do not absorb the tax once every common cofactor divisor has been removed. The separate corank-one cofactor/determinant construction for constructing such primitive kernels remains an unformalized derivation. The checked statement here is only the lcm divisibility after cofactor removal.

5 The prime unit translator

The channel tax raises an obvious question: is there any support that pays the tax at exactly one channel? There is, and it is explicit.

Theorem 5.1 (prime unit translator). *Let p be prime and take the coefficient-index pair $(p, -1)$ on the indices $(p-1, p)$. Then the factorial moment is 0; every channel $d < p$ vanishes, by the exact quotient identity $\lfloor (p-1)/d \rfloor = \lfloor p/d \rfloor$; every channel $d > p$ vanishes, because both indices lie below d ; and the p -channel numerator is exactly $p! - 1$.*

The proof holds uniformly for every prime. It supplies, at zero cost in the moment, a unit in the p -channel. Adding an integer multiple of the translator to any candidate kernel shifts the p -channel numerator by multiples of $p! - 1$ and leaves every other channel and the moment untouched.

The consequence is already uniform in the support location. For every channel rank and every prescribed cutoff, Lean constructs a factorial-grid kernel and a remote prime-translator pair entirely beyond that cutoff, with all requested low channels zero, nonzero factorial moment, and residual in $[-1/2, 1/2]$; see *exists remote factorial grid prime translator reduction*. The rounded residual may still equal zero, and strict nonvanishing has not been proved for any cofinal family. Even strict nonvanishing would only exclude denominators dividing the associated determinant; without a bound below the rational lattice spacing, it only excludes divisors of the associated determinant. Section 12 states the stronger scalar nonintegrality condition.

6 Cofinal strict-successor divisibility misses

6.1 Grid plateaux and first exit

A second approach uses the rational grid. Let H be a partial sum and q a candidate denominator. The kernel checks the algebraic grid threshold: writing $qH = k + r$ and $q(S - H) = u$, the next q^{-1} grid point $(k+1)/q$ lies below S exactly when $1 \leq r + u$. It also checks the factorial plateau theorem: if $H < G \leq S$, if $n!G$ is integral, and if $n!(S - H) < 1$, then the strict successor of $n!H$ and the canonical floor of $n!S$ are the same grid integer.

Two rigidity statements follow, both checked. Consecutive plateau floors, scaled by the next radix, force the canonical factorial digit to vanish. And any first-exit offset $\delta \in [0, 2)$ with carry $b = -\lfloor \delta \rfloor$ satisfies $b \in \{0, -1\}$: the exit is rigid, with exactly two alternatives.

The checked first-crossing argument continues from the exit to a denominator lower bound. For a rational grid level G , suppose that τ is its first crossing by the literal partial sums and write

$$G - H_{\tau-1} = \frac{a}{v}, \quad a, v > 0.$$

Then $v \geq \tau! - 1$. On the -1 exit branch this strengthens to $v \geq \tau!(\tau! - 1)$. No coprimality hypothesis on a and v is required.

There is also a direct obstruction at prime indices. Let

$$H_m = \sum_{2 \leq n \leq m} \frac{1}{n! - 1}, \quad Z_m = \lfloor m! H_m \rfloor + 1,$$

and let Δ_m be the distance from $(m-1)!H_{m-1}$ to its strict integer successor. The kernel checks, for $m \geq 3$, the exact criterion

$$m \mid Z_m \iff 1 + \frac{1}{m! - 1} < m\Delta_m \leq 2 + \frac{1}{m! - 1}.$$

If $S = a/q$ with $q > 0$, then for every prime $p > q$ the tail bound forces $p \mid Z_p$. Consequently, one exact missed prime p implies $q \geq p$. The rational implementation of Z_p agrees with the real-floor definition, and exact kernel reduction gives $11 \nmid Z_{11}$. Thus every rational representation of S has denominator at least 11.

6.2 The lossless carry criterion

The all-index recurrence is stronger. Define its exact carry by

$$Z_m = mZ_{m-1} + 1 - b_m.$$

Theorem 6.1 (exact carry characterization). *The following conditions are equivalent:*

$$S \notin \mathbb{Q}, \quad (\forall B)(\exists m > B) b_m \neq 1, \quad (\forall B)(\exists m > B) m \nmid Z_m.$$

Equivalently, S is rational precisely when $b_m = 1$ for all sufficiently large m .

If $S = a/q$ and $m-1 \geq q$, the plateau theorem identifies

$$Z_{m-1} = \frac{(m-1)!}{q} a, \quad Z_m = \frac{m!}{q} a = mZ_{m-1},$$

so necessarily $b_m = 1$. Hence one exact non-unit carry at index m forces $q \nmid (m-1)!$, and therefore also $q \geq m$. Conversely, if $b_m = 1$ eventually, then $Z_m/m!$ is eventually constant. The checked one-cell bound

$$H_m < \frac{Z_m}{m!} \leq H_m + \frac{1}{m!}$$

and the exact tail estimate show that $Z_m/m! \rightarrow S$; hence that eventual constant is S and is rational. Theorem 6.1 therefore reduces the original irrationality question to proving the existence of arbitrarily late non-unit carries. Exact rational normalization gives

$$60 \nmid Z_{60}, \quad 64 \nmid Z_{64}, \quad 67 \nmid Z_{67}.$$

At $m = 60$ the recurrence also proves $b_{60} \neq 1$, and hence $q \geq 60$. Since 67 is prime, the prime-miss theorem applied at 67 gives the stronger checked bound

$$S = \frac{a}{q}, q > 0 \implies q \geq 67.$$

There is also a sufficient arithmetic specialization at doubled prime indices. For every odd prime p , the kernel now specializes the strict-successor prime-power criterion to the literal prefixes:

$$p^2 \mid Z_{2p} \iff (b_{2p} = 1 \text{ and } p \mid Z_{2p-1}) \text{ or } (b_{2p} = 1 + p \text{ and } p \mid 2Z_{2p-1} - 1).$$

Consequently, failure of both displayed branches for a cofinal family of odd primes proves S irrational. This is a sharper two-stage target than a bare square nondivisibility assertion: it exposes separately the only two carry values and predecessor residues that can occur. The cofinal failure of both branches remains unproved.

A separate exact GMP integer computation certifies all 299998 carry cells for $3 \leq m \leq 300000$. Its unit carries occur exactly at

$$52, 591, 1030, 1407, 1438, 2164, 4258, 10991, 21236,$$

so no further unit carry occurs through $m = 300000$, where $b_{300000} \neq 1$. Applying the checked divisibility theorem gives

$$S = \frac{a}{q}, q > 0 \implies q \nmid 299999!.$$

In particular $q \geq 300000$, but the divisibility statement is stronger: it excludes every divisor of $299999!$, including divisors larger than 300000. It does not exclude all smooth denominators: a sufficiently high prime power can fail to divide $299999!$ even when its prime is small. The computation uses no floating-point arithmetic; its canonical data, Python driver, and GMP backend are hash-bound in the companion research packet. This remains a finite exclusion. Irrationality requires the same conclusion beyond every cutoff.

An independent continued-fraction calculation is recorded, with its source and receipt, in the research packet. At an 80000-bit exact bracket it forces 23449 common partial quotients and verifies the penultimate convergent's separation from S , yielding $q \geq 2^{39990} > 10^{12038}$ by best approximation of the second kind. This is also a finite exclusion. It bounds the size of q , whereas the carry calculation restricts its prime-power multiplicities.

6.3 The adjacent-unit window reduces to a universal bound

Consider an attempt to exclude two consecutive unit carries by extracting a prime-power or quotient-gcd obstruction from their cleared width-one window. The exact two-step recurrence reduces this proposed obstruction to a universal bound. Let U_k and V_k be the positive numerator and denominator of the reduced predecessor gap, and let $G_k > 0$ be the exact transition normalizer. For $m \geq 3$, the two unit carries are equivalent to an integer offset Ω_m satisfying

$$0 < \Omega_m \leq D_m.$$

The window denominator telescopes independently of the carry values:

$$D_m = V_{m+2}G_{m+1}G_m = V_m(m! - 1)((m+1)! - 1).$$

More generally, before imposing either carry value, the cleared offset has the exact mixed-radix factorisation

$$\Omega_m = U_{m+2}G_{m+1}G_m + D_m((m+1)b_m + b_{m+1} - (m+2)).$$

The second summand records the complete two-carry defect. This identity identifies the Archimedean window and the reduced-denominator dynamics as two descriptions of the same integer. Under the adjacent-unit assumption, the offset has the matching factorisation

$$\Omega_m = U_{m+2} G_{m+1} G_m.$$

Cancelling the common positive normalizer therefore reduces the complete window to

$$0 < U_{m+2} \leq V_{m+2},$$

which is the universal bound for a reduced strict-successor gap.

Consequently, any prime, prime-power, or quotient-gcd obstruction extracted only after assuming the two carries is just the universal reduced-fraction bound rewritten through a common factor. To exclude adjacent unit carries, one must instead constrain the predecessor state before making that assumption, for example by Archimedean non-concentration or independent prefix arithmetic.

6.4 Why nearest-integer nonvanishing is only a denominator screen

The nearest-integer residual used by a rank-two channel construction has a precise rational meaning, but it is not by itself a small-linear-form contradiction. Write a hypothetical value as $S = a/q$ in lowest terms and, for integers N, K , put

$$\delta = NS + K - \text{nint}(NS + K), \quad -\frac{1}{2} \leq \delta \leq \frac{1}{2}.$$

Then $q\delta$ is integral. Consequently

$$\delta = 0 \iff q \mid N, \quad \delta \neq 0 \implies |\delta| \geq \frac{1}{q}.$$

The universal nearest-integer bound therefore remains perfectly compatible with rationality: for $S = 1/q$ and $N = 1$, the residual is nonzero while it still lies in $[-1/2, 1/2]$. Its strict nonvanishing can only screen the denominator through $q \nmid N$; it supplies no irrationality conclusion without a stronger, denominator-sensitive comparison.

For the factorial-gap rank-two construction, the checked Cramer identity identifies N with the appended determinant $\det A$, while the prime translator changes only an integral coordinate. Thus a nonzero rounded residual says exactly that a hypothetical denominator does not divide $\det A$. To prove irrationality by this method, one would need such nonvanishing cofinally, together with either a bound below the rational lattice spacing or an independent contradiction for every possible denominator. Neither assertion is proved here.

There is also a checked finite peeling identity. For $x \neq 0, 1$ and $K \geq 0$,

$$\frac{1}{x-1} = \sum_{j=1}^K \frac{1}{x^j} + \frac{1}{x^K(x-1)}.$$

When $x = k!$ and a chosen factorial scale is divisible by $(k!)^K$, the scaled finite sum is integral and only the last term retains the factor $k! - 1$ in its denominator. The identity isolates one residual fraction before exact bounding; it does not yet give a cofinal family of nonzero residuals.

One proposed strengthening is false and is recorded as such: the divisibility $(m! - 1) \mid \text{den}(V_m)$ fails at the reported strict events $m = 52$ and $m = 591$. Only the Archimedean first-crossing lower bound survives.

The carry equivalence states the full remaining problem. Wilson factors, doubled-prime branches, congruence tests, nearest-integer residuals, and the finite GMP computation give ways to study individual indices or exclude denominators. Irrationality still requires the quantifier $(\forall B)(\exists m > B)$.

7 Finite prime-pole cancellation

The lcm of a finite prefix records the largest prime-power pole, but it does not say whether that pole survives when the summands are added. This cancellation is now exact. Put

$$L_M = \text{lcm}_{2 \leq n \leq M} (n! - 1), \quad A_M = \sum_{2 \leq n \leq M} \frac{L_M}{n! - 1},$$

so that $H_M = A_M/L_M$ before reduction. Fix a prime q , and suppose $e \geq 1$ is the largest exponent for which q^e divides at least one displayed denominator. Let

$$I_{q,e}(M) = \{n \in [2, M] : v_q(n! - 1) = e\}$$

and define its principal residue

$$R_{q,e}(M) = \sum_{n \in I_{q,e}(M)} \left(\frac{n! - 1}{q^e} \right)^{-1} \pmod{q}.$$

All terms of lower q -adic order vanish after multiplication by L_M and reduction modulo q . Lean proves the surviving identity

$$A_M \equiv \frac{L_M}{q^e} R_{q,e}(M) \pmod{q}. \quad (6.1)$$

Maximality of e makes L_M/q^e a unit modulo q . Consequently

$$v_q(\text{den } H_M) = e \iff R_{q,e}(M) \not\equiv 0 \pmod{q}. \quad (6.2)$$

This is the exact finite prime-pole criterion: support and multiplicity are compressed into one residue-weighted sum, with no loss between the displayed factorial gaps and the reduced prefix.

The distinction is arithmetically real. At $M = q - 1$, Wilson's theorem always provides the endpoint hit $q \mid (q - 2)! - 1$, but it does not isolate that hit or prevent cancellation at the maximal layer. The first exact cancellation found by the local scan is

$$q = 139, \quad I_{139,1}(138) = \{69, 122, 137\},$$

with maximal-hit cofactors 6, 49, 73 and

$$6^{-1} + 49^{-1} + 73^{-1} \equiv 0 \pmod{139}.$$

The second is

$$q = 2593, \quad I_{2593,1}(2592) = \{349, 2243, 2591\},$$

with cofactors 1508, 1566, 1678 and reciprocal sum zero modulo 2593. Lean checks both displayed reciprocal equalities by exact computation. A separate local modular scan through $q \leq 3000$ reproduces precisely these two examples and all the displayed data. The larger external scan through 200000 has not been locally regenerated and is not used as theorem authority here.

Equations (6.1)–(6.2) have two immediate consequences. First, a large prime factor or Wilson endpoint hit is not automatically a pole of the reduced prefix; one must control the entire maximal-hit residue. Singleton maximal support is the easy noncancelling case, since its cofactor is a unit. Second, incidence bounds that count hits but ignore their inverse cofactor weights can at best control the size of $I_{q,e}(M)$, not the cancellation that decides survival. A proof along these lines would need a cofinal family of primes and prefixes for which this weighted residue is nonzero, followed by an independent Archimedean argument linking the surviving pole to the strict-successor interval. Equations (6.1)–(6.2) settle the local arithmetic part of that problem.

8 A four-step defect escape criterion

The following recurrence gives another necessary condition. Let e_n and η_n be integer sequences related by

$$\eta_n = (n+1)e_n - e_{n+1}, \quad Q(n) = n^2 + 2.$$

Lean checks the following four-step alternative. If $e_n > 0$ and $\eta_{n+j} < n+j$ for $j = 0, 1, 2, 3$, then

$$e_{n+4} > Q(n+4).$$

Applying the same calculation to $-e$ and $-\eta$ gives the signed form: if $e_n \neq 0$ and $|\eta_{n+j}| < n+j$ through the same four positions, then $|e_{n+4}| > Q(n+4)$. Consequently an eventual quadratic window $|e_n| \leq Q(n)$ together with eventual sublinear defects forces $e_n = 0$ eventually. In the one-sided factorial-neighbour specialization, cofinally many nonzero errors inside that window force cofinally many indices with $\eta_k \geq k$.

The proof is elementary but the integer gap matters. From $e_n \geq 1$, rewrite $e_{r+1} = (r+1)e_r - \eta_r$ four times and insert the largest permitted defect at each step. The resulting integral lower bound lies beyond the quadratic window. Sign normalization supplies the two-sided statement.

The checked recurrence quantifies the growth forced on a factorial-neighbour approximation. Applying it to the present series would require a reciprocal weighted transform connecting these errors to S , a proof of the stated window for that transform, and cofinal defects for the actual strict successors. None of these three inputs is presently available. The open step remains to prove $m \nmid Z_m$ beyond every cutoff.

9 Weighted projection rigidity

The third checked layer converts modular disagreement into exclusion.

Theorem 9.1 (projection rigidity). *Suppose the natural endpoint numerator Z is congruent to a weighted numerator T modulo R . Then every divisor Q of R with $Z \leq B < Q$ satisfies $T \bmod Q = Z$. Consequently two divisors $Q_1, Q_2 > B$ with unequal projected residues exclude any such bounded endpoint; and unequal projections force $\min(Q_1, Q_2) \leq T$.*

The leave-one-out specialisation $Q = R/r$ is also checked. More generally, let a, b divide R and take the complementary projection moduli R/a and R/b . Lean checks the same quotient cancellation and collision-cap comparison for these factor projections. If $\gcd(a, b) = 1$, then $\text{lcm}(R/a, R/b) = R$, and the resulting branch-free factor-pair floor is at most the global complementary residue. The factors a, b may both divide one private quotient. Thus the checked reduction needs no analytic input and no pair of distinct denominator indices, only suitable factors whose projections or factor-pair floor satisfy the stated bound. This is checked in *factorial block factor projection lcm eq private modulus* and *factorial block complementary factor pair floor le global*.

Lean also verifies the construction for the literal factorial block. It builds the collision core C , private quotients r_i , private modulus R , and weighted numerator T for the actual denominators $i! - 1$, and proves both the endpoint congruence modulo R and the required coprimality. Moreover, if $m! - 1$ has canonical large prefix-private primes, then their complete prime-power product divides the single quotient owned by m on the tailored block with parameter $\lfloor m/2 \rfloor + 1$, hence divides that block's R .

The collision core itself now has an exact incremental law. For the positive factorial-gap denominators, adjoining d_a to an old finite family S gives

$$C(S \cup \{a\}) = \text{lcm}(C(S), \gcd(d_a, \text{lcm}_{j \in S} d_j)).$$

Indeed, finite-family gcd-lcm distributivity collapses the lcm of all pairwise gcds against d_a to this single gcd. The same formula holds after adjoining the distinguished base; see *pairwise*

collision core insert gcd lcm and *collision core insert gcd lcm*. Thus each step needs only the old denominator lcm and the old collision core, with no pairwise rescan.

There is now also an exact product-lcm bound. If $\tilde{C}(S)$ denotes the collision core after cancelling a positive distinguished base, while $L(S) = \text{lcm}_{j \in S} d_j$ and $P(S) = \prod_{j \in S} d_j$, then Lean proves

$$\tilde{C}(S)L(S) \mid P(S), \quad \text{hence} \quad \tilde{C}(S) \leq \frac{P(S)}{L(S)}.$$

See *collision core div base mul denominator lcm dvd denominator prod*. For the actual factorial block this specializes to

$$\text{factorialBlockNormalizedCollisionCore}(p) \leq \frac{\prod_{n \in I_p} (n! - 1)}{\text{lcm}_{n \in I_p} (n! - 1)},$$

where I_p is the block index set; see *factorial block normalized collision core le gap prod div gap lcm*. This inequality converts lower estimates for the factorial-gap lcm into upper estimates for the normalized collision core. Its use in an irrationality proof would still require cofinal estimates strong enough at the selected private factor and factorial scale. A fixed-modulus hit count alone does not supply such control.

The distinguished-base cancellation is now exact prime by prime. Writing $B = (p - 1)!$ and C for the unnormalised factorial-block collision core,

$$\tilde{C}_p = \frac{\text{lcm}(B, C)}{B} = \frac{C}{\text{gcd}(B, C)}, \quad v_q(\tilde{C}_p) = v_q(C) - \min\{v_q(B), v_q(C)\}.$$

See *collision core div base eq pairwise collision core div gcd*, *collision core div base factorization*, and *factorial block normalized collision core factorization*. Consequently $q^e \mid \tilde{C}_p$ exactly when the pairwise core carries $q^{e+v_q(B)}$; in the factorial block this forces two distinct gaps to be divisible by that higher power. Lean moreover proves the sharp surviving valuation cap

$$v_q(\tilde{C}_p) + v_q((p - 1)!) < q.$$

Thus every support prime satisfies $p - 1 < q(q - 1) < q^2$, and, whenever $k(k - 1) \leq p - 1$, $\tilde{C}_p$ is coprime to $k!$; see *factorial block normalized collision factorization add base lt prime*, *factorial block prime sq gt pred of dvd normalized collision core*, and *factorial block normalized collision core coprime factorial of mul pred le*. This removes every factorial channel below the moving square-root cutoff. The aggregate product of the remaining large prime powers at the selected quotient, as well as the complementary projections and residues, still needs cofinal control.

For collision estimates that already provide an upper-half hit, no exponent is lost to normalization. If q divides a displayed factorial gap at some $n \geq p$, then $q \nmid (p - 1)!$, and Lean proves for every $e > 0$ that

$$q^e \mid \tilde{C}_p \iff q^e \mid C_p.$$

See *factorial block prime not dvd base of upper hit* and *factorial block prime power dvd normalized collision core iff of upper hit*. Combined with the two-hit theorem, this identifies every complete normalized upper-hit contribution with repeated full-power load in two distinct displayed gaps. The remaining arithmetic task is to aggregate those moving loads strongly enough for the normalized collision cap; this equivalence does not provide that estimate or the complementary-residue bound.

This bridge has an exact incidence-count form. For an upper-hit prime q and every $e > 0$, Lean proves

$$q^e \mid \tilde{C}_p \iff 1 < \#\{i \in I_p : q^e \mid i! - 1\}.$$

See *factorial block prime power dvd normalized collision core iff one lt hit count*. Hence a source estimate giving at most one q^e -hit deletes that exponent from the normalized core and yields $v_q(\tilde{C}_p) < e$; see *factorial block normalized collision core factorization lt of hit count le one*. The remaining problem is genuinely aggregate: obtain sufficiently uniform incidence bounds over all moving support primes and exponents, multiply the surviving valuation contributions, and still close the complementary-residue coordinate.

The local aggregation is now exact. For every upper-hit prime q , Lean proves

$$v_q(\tilde{C}_p) = \# \{e \in [1, q-1] : 1 < \#\{i \in I_p : q^e \mid i! - 1\}\};$$

see *factorial block normalized collision core factorization eq repeated hit layer count*. There is therefore no additional valuation loss between prime-power incidence estimates and the complete local collision exponent. The open step is to bound these layer counts uniformly as q and p move, then control the product over all surviving primes strongly enough for the normalized collision cap; this theorem does not supply that global estimate.

The same local load now has a distance-sensitive witness. Put $B = (p-1)!$. If q is prime, $e > 0$, and $q^e \mid \tilde{C}_p$, Lean produces $i < j$ in I_p such that

$$q^{e+v_q(B)} \mid i! - 1, \quad q^{e+v_q(B)} \mid j! - 1, \quad q^{e+v_q(B)} \leq j^{j-i}.$$

See *exists factorial block hit pair with normalized prime power le gap pow*. Consequently, if

$$(2p-1)^d < q^{e+v_q(B)},$$

then some such two hits satisfy $d < j-i$; see *exists factorial block hit pair distance gt of endpoint pow lt normalized prime power*. The spacing hypothesis in that reduction is now discharged internally. If q is prime, then any two q^e -hits $i < j$ satisfy $e < j-i$, without an endpoint or large-prime hypothesis; see *factorial block prime hit pair distance gt exponent*. The point is that $q \mid j! - 1$ already forces $j < q$, while the preceding gap-power inequality converts this automatic size relation into strict separation. Consequently Lean proves the global primewise diameter ceiling

$$q \mid \tilde{C}_p \implies v_q(\tilde{C}_p) + v_q((p-1)!) < 2p-3$$

for every prime q and $p \geq 2$; see *factorial block normalized collision factorization add base lt block diameter*. The exponent-level version is *factorial block normalized collision exponent add base factorization lt block diameter*. Thus the earlier endpoint-prime estimate is a special case, and even primes already present in the normalization base pay for their base valuation inside the same block-diameter budget. This still does not control how many collision primes occur or the product of their bounded powers; those global estimates, together with the complementary-residue bound, remain open.

The pairwise statement is stronger than the selected-witness form used in that proof. For arbitrary q, e and any displayed hits $i < j$, Lean proves

$$q^e \mid (i! - 1), \quad q^e \mid (j! - 1) \implies q^e \leq j^{j-i};$$

see *factorial block prime power le gap pow of two hits*. Consequently, when q is prime and $e > 0$, every two q^e -hits in the block—not just one chosen pair—satisfy $e < j-i$; see *factorial block prime hit pair distance gt exponent*. Every prime-power hit layer is therefore an e -separated subset of the block. Lean now proves the finite cardinality corollary itself:

$$(e+1)\#\{i \in I_p : q^e \mid i! - 1\} \leq 2p + e - 2$$

for $p \geq 2$, prime q , and $e > 0$; see *factorial block prime power hit count mul succ le*. The remaining task is to combine this packing estimate with the exact repeated-layer valuation identity, sum

the prime-power weights over all moving collision primes, and prove a global product bound strong enough for the normalized collision cap.

For an endpoint prime carrying one upper-half hit, Lean now performs the first combination exactly. If $p \geq 2$, q is prime, and $2p - 1 < q$, then

$$v_q(\tilde{C}_p) = \# \{e \in [1, 2p - 4] : 1 < \#\{i \in I_p : q^e \mid i! - 1\}\};$$

see *factorial block normalized collision core factorization eq truncated repeated hit layer count*. Thus every repeated-hit layer outside the block-diameter window has been removed from the exact local valuation formula. The remaining estimate is still global and weighted: these truncated layer counts must be aggregated over the moving endpoint primes strongly enough to bound their complete prime-power product, and the independent complementary-residue coordinate remains open.

The endpoint incidence criterion itself no longer needs a selected upper-half anchor. For every prime $q > 2p - 1$ and $e > 0$, Lean proves

$$q^e \mid \tilde{C}_p \iff 1 < \#\{i \in I_p : q^e \mid i! - 1\};$$

see *factorial block prime power dvd normalized collision core iff one lt hit count of endpoint lt base*. Thus an at-most-one q^e incidence estimate forces $v_q(\tilde{C}_p) < e$ without first choosing an upper hit; see *factorial block normalized collision core factorization lt of endpoint lt base of hit count le one*. At $e = 2$ this gives the conditional squarefree conclusion $v_q(\tilde{C}_p) \leq 1$; see *factorial block normalized collision core factorization le one of endpoint lt base of prime sq hit count le one*. More generally the endpoint inequality can be replaced by the exact condition $q \nmid (p - 1)!$. For every such prime and every $e > 0$, Lean proves the same hit-count equivalence; see *factorial block prime power dvd normalized collision core iff one lt hit count of not dvd base*. The at-most-one hypothesis cuts the normalized valuation below e , and for $e = 2$ it gives conditional squarefreeness; see *factorial block normalized collision core factorization lt of not dvd base of hit count le one* and *factorial block normalized collision core factorization le one of not dvd base of prime sq hit count le one*. Every prime $q \geq p$ is absent from $(p - 1)!$, so this covers the entire moving prime range at and above the block parameter. The missing inputs are a uniform prime-square incidence theorem and a global weighted product estimate. The squarefreeness premise is not proved. An exhaustive modular scan through $q \leq 2,000,000$ and $n \leq 240$ found four individual square hits and no prime with two such hits. Separately, all 498,501 pairs $2 \leq a < b \leq 1000$ have squarefree $\gcd(a! - 1, b! - 1)$, and the aggregate squarefree-collision scan through $p = 499$ stays below 0.374 of the upper-descending-factorial logarithmic scale. These are finite exact computations. They do not provide an asymptotic incidence bound.

Cofinal prefix-private support itself is unconditional. Given any cutoff B , Lean chooses a prime $q \geq B! + 5$, uses Wilson's theorem to obtain $q \mid (q - 2)! - 1$, and takes the least factorial-gap hit m of q . If $m \leq B$, then $q \leq m! - 1 \leq B!$, a contradiction. Hence $m > B$; see *cofinal prefix private factorial gap hits*. A finite variant compares the product of a chosen set of primes, each at least 5, with

$$\prod_{2 \leq k \leq B} (k! - 1).$$

If the prime product is larger, at least one chosen prime has no hit through B , while Wilson still bounds its least hit by $q - 2$; see *exists late prefix private factorial gap hit of prime product lt*. These statements supply private factors, but they do not prove either scale estimate below. In particular, the unconditional construction gives no useful upper bound for q in terms of its least hit m .

Wilson reflection also limits what can be inferred from a prime factor merely because it is linear in a later index. If n is odd, $n < q$, and $q \mid n! - 1$, then $q \mid (q - n - 1)! - 1$. When both indices lie in the same block and the reflected hit is earlier, equivalently $q < 2n + 1$, this repeated hit survives predecessor-factorial normalization and its full-block incidence count exceeds one; see *prime dvd reflected factorial gap of odd* and *prime dvd factorial block normalized collision core of odd*.

reflection and one l t factorial block prime hit count of odd reflection. Thus a linear-size divisor need not be private.

The point follows directly from Stewart's stated theorem. For every $\varepsilon > 0$ there are infinitely many odd n whose least prime factor q of $n! - 1$ satisfies

$$n < q < \left(\frac{\sqrt{145} - 1}{8} + \varepsilon \right) n;$$

the printed text explicitly transfers estimate (9) from $n! + 1$ to $n! - 1$ [4, p. 464]. Wilson reflection then supplies the earlier hit $q \mid (q - n - 1)! - 1$. The source controls q relative to the later index n , but it does not control q relative to the private first-hit index m . Accordingly it contributes to the collision-core estimate; it supplies neither a private-anchor estimate nor a global product estimate.

The strongest checked factor-level consumers can be stated without the earlier two-owner architecture.

Theorem 9.2 (moving private-factor scale split). *For every cutoff B , suppose there are $m \geq 4$ and a canonical large prefix-private prime $q \mid m! - 1$ such that, for $p = \lfloor m/2 \rfloor + 1 > B$,*

$$(2p + 1)L_p < 2p^2(2p - 1)!\rho_p, \quad (2p + 1)C_p q < 2p^2(2p - 1)!.$$

Then S is irrational.

Theorem 9.3 (split-factor normalized-collision criterion). *Suppose that cofinally many prime block parameters p admit factors $a, b \mid R_p$, with $R_p > 1$, whose complementary projections modulo R_p/a and R_p/b disagree and for which*

$$(2p + 1)\tilde{C}_p \max\{a, b\} < 2p^2(2p - 1)!.$$

Then S is irrational. The factors a and b may lie in one moving private owner quotient; they need not come from distinct denominator indices.

In fact one selected prime q already furnishes the exact coprime factor pair $(1, q)$: its projection moduli are R and R/q , whose least common multiple is R . Thus no second selected prime is needed; see *factorial gap large prefix private power modulus dvd tailored block private quotient* and *factorial gap large prefix private primes unit factor pair tailored block*. There is no hidden equality/disagreement branch in this specialization. Writing ρ for the global complementary residue, Lean proves that the unit-pair floor is exactly

$$\min\{\rho, R/q\};$$

see *factorial block unit factor pair floor eq min*. Consequently the remaining factor-pair scale comparison must simultaneously beat the global complementary-residue coordinate and the local R/q coordinate. After using $L = CR$, the latter is precisely the collision-cap comparison with the selected factor q , while the former is the global complementary-residue lower bound. Lean records this as the exact equivalence

$$(2p + 1)L < 2p^2(2p - 1)!\min\{\rho, R/q\} \iff \begin{cases} (2p + 1)L < 2p^2(2p - 1)!\rho, \\ (2p + 1)Cq < 2p^2(2p - 1)!, \end{cases}$$

see *factorial block unit factor pair floor scale iff*. Thus the unit-factor reduction has no opaque floor premise left: the two surviving arithmetic estimates are exposed independently and neither follows merely from the existence of the selected prime. The resulting irrationality criterion works

for every natural block parameter at least three, including composite parameters. Wilson's theorem supplies cofinal prefix-private factors without analytic input. The stronger source-backed large-prime selection supplies a positive-density family and a linear lower bound for q relative to the original hit. To apply the criterion, one must still prove both the global complementary-residue bound and the local collision-core bound cofinally. These two requirements are packaged by *irrational factorial gap series of cofinal large prefix private unit scale split*.

10 Obstructions to several proposed arguments

The following calculations show why several natural arguments cannot prove irrationality in their present form.

- Residue vectors, their recurrences, and window widths admit synthetic all-hit blocks. They cannot prove irrationality on their own.
- Known pointwise prime congruences, prime-dilation congruences, parity, and the exact prime coefficient formula admit a synthetic rational countermodel. A congruence family that a rational number could also satisfy decides nothing.
- Wilson quotients, harmonic sums, p -adic gamma identities, and factorial residues do not control the required Archimedean floor without an additional coupling theorem. Every prime-window test factors into a sharp Archimedean strict-ceiling condition and a modular divisibility condition, and the missing ingredient is a theorem coupling the Archimedean and modular conditions.
- Wilson-endpoint support or hit-count bounds alone do not prove that a prime power survives reduction of a factorial-gap prefix. The principal-residue formula of §7 is exact, and the scan's cancellations at $q = 139$ and $q = 2593$ show that the reciprocal cofactor weights are indispensable. Singleton maximal support, by contrast, cannot cancel.
- Fixed-denominator scalar canonical-product localisers and rank-saturated consecutive-jet Hermite–Padé systems pay the full factorial-gap denominator. For $E(z) = \prod_{n \geq 2} (1 - z/n!)$ the genus-zero product satisfies $-E'(1)/E(1) = S$, and the natural scalar linear form carries the coefficient $Q_N = \prod_{2 \leq n \leq N} (n! - 1)$, for which Q_N times the tail diverges. Exact first-order interpolation, scalar residue weighting, the natural Wronskian, and rank-saturated consecutive jets all reassemble the same prohibitive denominator.
- The support-matroid layer cannot create an additional fractional cancellation coordinate (§4), and factorial valuations cannot absorb the channel LCM tax.
- A fixed pair of low-index private owners cannot yield a cofinal projection argument. If the owner index n is fixed and $p > n! - 1$, then $n! - 1 \mid (p - 1)!$, so its private quotient in the factorial block at p is exactly one. Lean checks this uniformly in *factorial block private quotient eq one of gap lt* and checks the two-owner consequence in *factorial block fixed pair private quotients eq one*. Thus the large private quotients seen at small blocks, for example the factor 719 owned at $n = 6$, are finite-range phenomena. The factor-level reduction does not require two moving denominator indices: two factors inside one moving private quotient can suffice. It still requires selected nontrivial factors that escape with p .

Proposition 10.1 (fixed-owner absorption boundary). *Fix denominator indices $i, j \geq 2$. Whenever*

$$p > \max\{i! - 1, j! - 1\},$$

both corresponding private quotients in the factorial block at p equal one. Hence any cofinal private-factor certificate must use owner indices or prime factors that escape with the block parameter.

11 Finite certificates

The following are computations. Each excludes exactly the denominators it names and nothing more.

The finite-support vector $\lambda = 2e_3 - e_4$ has, by kernel check, $V_2 = 0$, factorial moment -12 , $V_3 = -2$, $V_4 = 11$, and $V_d = -12$ for every $d \geq 5$. Under the exact rational tail enclosure $1/119 < \Theta_4 < 1/50$, its residual lies strictly between $-93/575$ and $-309/13685$; in particular it is nonzero and subunit.

Exact integer regeneration verifies the canonical primitive kernels for every $2 \leq D \leq 12$: channels 2 through D vanish, the factorial moment is L_D , and the coefficient content is one. At $D = 9$ the moment is $L_9 = 31540008254514077395$ and, after the stated prime-unit shift, $1353/100000 < R_9 < 1354/100000$. At $D = 3$ the vector $c = (-40, 55, -10, 1)$ on the support $(3, 4, 5, 6)$ annihilates channels 2 and 3, has moment 600, and satisfies

$$0.09925341997208298 < L_3(c) < 0.09925341997208300.$$

This excludes denominators dividing 600.

There are two different computations through the same range. An external interval computation reports the stronger geometric statement that no zero-branch event occurs at any $m \leq 100000$; its cited executable and source digest were not supplied, so that zero-branch classification remains external finite evidence. The strict-successor carry computation used above is local and independently regenerated: its exact source, GMP backend, canonical data, and receipt digests are recorded in the research packet. The two claims must not be conflated. The local certificate establishes $b_{300000} \neq 1$, and the checked theorem converts precisely that fact into $q \nmid 299999!$ (and hence $q \geq 300000$). The bracket-certified continued-fraction prefix independently raises the finite size exclusion to $q \geq 2^{39990} > 10^{12038}$; its script and receipt are named above, and it likewise does not change a quantifier.

All of these conclusions are finite.

12 Complements and further questions

Four questions remain.

On the channel side, strict nonvanishing by itself is only a denominator screen. The uniform remote-support theorem of §5 already produces the low-channel kernel, a nonzero moment, and a residual in $[-1/2, 1/2]$ beyond every cutoff. For a hypothetical reduced denominator q , a nonzero residual only says that q does not divide the associated determinant; the bound $1/2$ does not put it below the rational lattice spacing $1/q$. The scalar question is therefore the following: for a hypothetical reduced denominator b , for some D and some nonzero m divisible by $\text{lcm}(b, L_D)$ with mT_D nonintegral, equivalently for k with $0 < |mT_D + k| < 1$. The formerly proposed determinantal inequality $0 < |N(I, D)| < \Delta(I, D)$ is impossible: Laplace expansion along the appended row gives $\Delta(I, D) \mid N(I, D)$, so nonzero N has $|N| \geq \Delta$. This divisibility rules out the proposed inequality for every choice of I and D .

On the projection side, endpoint transport to the literal factorial block is checked. Cofinal prefix-private factor supply is also available: Wilson gives an unconditional existence theorem, while the source-backed large-prime theorem and least-hit bridge give the stronger canonical selection. What is not known is a cofinal family for which the unit pair $(1, q)$ satisfies both the global complementary-residue bound and the local collision-core bound. The factor cannot stabilise at fixed low-index data: the absorption theorem above makes every such private quotient equal to one for large p . Reflection further shows that a linear-size factor at an odd later index may be a repeated hit in the collision core rather than private support.

On the prime-pole side, the exact local target is now $R_{q,e}(M) \neq 0 \pmod{q}$ for a suitably controlled cofinal family. Mere existence of a large divisor or a bounded number of maximal

hits is insufficient, as the two exact three-hit cancellations show; their reciprocal equalities are Lean checked. Even nonvanishing must then be coupled to the Archimedean strict-successor window; it cannot by itself change the global rationality quantifier.

On the grid side, the exact target is to prove that $m \nmid Z_m$ for arbitrarily large m (equivalently $b_m \neq 1$ cofinally). The adjacent-unit window of §6.3 cannot supply this after the pair assumption; an argument must constrain the predecessor state before that assumption. Prime-index subtargets remain useful: prove that infinitely many primes p fail $p \mid N_p$, or that for some fixed k infinitely many primes p fail $p^k \mid N_{kp}$, where N_m is the least integer strictly above $m!$ times the m -th partial sum; prove for cofinally many primes p that the least $((p-1)!)^{-1}$ grid point strictly above H_p is not S ; and beyond the finite range already discharged by $q \nmid 299999!$, control cancellation in the first-crossing gap V_p strongly enough to contradict $\text{den}(V_p) \geq \tau_p! - 1$. The prime-grid statement through $p \leq 300000$ is a corollary of the finite carry certificate.

Two further questions are structural rather than immediate. Find a global invariant of the true divisor-factorial coefficients that rules out eventual $q_m = 1$ and is absent from the synthetic congruence countermodels; the countermodels of §10 exist precisely because no such invariant is yet in hand. And construct or rule out an N -dependent, denominator-resonant, strict-subrank vector Hermite–Padé system that does not factor through scalar evaluation at $z = 1$ and does not reassemble the full factorial-gap lcm. The available calculations rule out the scalar and rank-saturated cases. The strict-subrank vector-valued case remains open.

Erdős #68 remains open: no statement above proves irrationality or excludes every rational value. The finite checked consequences are nevertheless unconditional: every rational representation has $q \geq 2^{39990} > 10^{12038}$ by the bracket-certified continued-fraction screen; the independent carry certificate separately supplies $q \nmid 299999!$.

Statements and declarations

The Lean declarations cited below are the proof authority for the formalized statements. The checked core comprises the canonical factorial digit kernel, the finite defect automaton algebra, floor-factorial channel arithmetic, the channel congruence and its integral normal form, the prime unit translator, weighted projection rigidity, the factor-split projection reduction, the fixed-index factorial-base absorption no-go, the rational-grid plateau and first-exit results, the first-crossing denominator bounds, and the literal-prefix prime obstruction through the exact $p = 11$ instance, strengthened by the all-index eventual-unit-carry theorem and the exact reductions at $m = 60, 64, 67$, the bound $q \geq 67$, and the finite geometric peeling identity. It also checks the adjacent-unit width-one criterion, the two-step denominator telescope and offset factorisation, and their reduction to the universal numerator bound. It checks the normalized strict-successor step and its finite factorial-series expansion in the carry defects $1 - b_m$, the convergence $Z_m/m! \rightarrow S$, and the exact equivalence between irrationality and cofinally many non-unit carries. The source-current Comparator front-loads the fixed companion orbit as `companionOrbit_completeCharacterization`, which repeats Theorem 2.1 and Equation (2) literally. Its second theorem packages the complete four-way strict-successor result, namely eventual unit carry, cofinal non-unit carry, pointwise carry/divisibility equivalence, and cofinal strict-successor divisibility failure, as `strictSuccessorCarry_completeCharacterization`, routed to Theorem 6.1. The finite second-layer decomposition, its digit identity $d_m^{(2)} = m - 1 - b_m$, the two-scale floor-stability theorem, and the eventually-maximal rational-digit exclusion are checked separately in `SecondLayerDigit.lean`. The maximal prime-pole numerator formula and the explicit reciprocal equalities at 139 and 2593 are checked in `PrimePoleCriterion.lean`; the exact scan supplies the maximal-hit data and resulting full cancellations. These two source-current modules sharpen the finite calculations, but neither proves a cofinal condition. The exact GMP carry certificate through $m = 300000$ is separately regenerated and hash-bound; combined with the

checked carry theorem it gives $q \nmid 299999!$ (and hence $q \geq 300000$), but it is not itself a Lean evaluation. The certified continued-fraction script and receipt provide the independent finite bound $q \geq 2^{39990} > 10^{12038}$. This exact bracket computation is independent of Lean and is finite. The digit-rationality equivalence, the weighted primitive support decomposition, and the determinant-quotient reduction are returned derivations that have not been kernel-checked here, and are labelled as such wherever they appear. The factorial-gap lcm growth bound is derived here from the exact factorial-congruence multiplicity theorem of Garaev–Luca–Shparlinski [3, arXiv v1, Thm. 12, p. 16]. We derive the bound from their theorem; it has not been formalized in Lean and no preceding result depends on it.

A Guide to the formal sources

The public `ErdosProblems.Erdos68` package contains the checked source for this note. The pinned release snapshot used by the declaration links contains the following core modules: `CanonicalFactorialDigits`, `ChannelBreakpointRigidity`, `ChannelIntegralCongruence`, `CompanionOrbitRationality`, `EndpointWeightedPrivateSupport`, `FactorialCarry`, `FactorialChannelCertificate`, `FactorialZeroPlateau`, `FiniteDefectAutomaton`, `PrimeUnitTranslator`, `PrimeZeroBranch`, and `StrictSuccessorArithmetic`. The release root imports every cited module. The declaration table below is pinned to the shared formal-source commit used throughout this problem-note series.

- [strict successor](#)
- [strict successor sub int](#)
- [canonical factorial digit](#)
- [factorial scaled rational eq int cast](#)
- [canonical factorial digit eq zero of rational](#)
- [prime pow dvd factorial dilation](#)
- [prime pow dvd factorial dilation div](#)
- [prime pow dvd cleared rational numerator](#)
- [residual centre term](#)
- [factorial coeff term](#)
- [residual centre](#)
- [factorial coeff rat](#)
- [factorial eq mul pred factorial](#)
- [pred div eq of not dvd](#)
- [pred div add one eq of dvd](#)
- [residual centre term of not dvd](#)
- [residual centre term of dvd](#)
- [residual centre term self](#)
- [factorial coeff term self](#)
- [residual centre recurrence](#)
- [factorial pow floor dvd factorial](#)
- [channel weight](#)

- [channel weight mul denominator](#)
- [channel numerator](#)
- [factorial moment](#)
- [channel event](#)
- [channel event eq zero of not dvd](#)
- [lambda34](#)
- [lambda34 apply three](#)
- [lambda34 apply four](#)
- [lambda34 channel two](#)
- [lambda34 factorial moment](#)
- [lambda34 channel three](#)
- [lambda34 channel four](#)
- [lambda34 channel ge five](#)
- [lambda34 residual of tail](#)
- [lambda34 residual bounds](#)

Source-current adjacent-unit no-go. Section 6.3 is checked in the local module `ErdosProblems/Erdos68/AdjacentUnitCarryWindow.lean`, principally through `consecutive_unit_carries_iff_positive_offset_le_den`, `twoStep_den_mul_transitionNormalizers`, `adjacentUnitCarryWindowDen_eq_twoStep_den`, and `adjacentUnitCarryWindowOffset_eq_twoStep_factorization`. The universal numerator bounds are in `PrimeZeroBranch.lean`. This module lies outside the older public declaration-table pin and must accompany the revised paper at its next public checkpoint.

Source-current second-layer and prime-pole modules. The finite digit law and two-scale floor-stability boundary are checked in `SecondLayerDigit.lean`; the general prime-pole formula and the 139 and 2593 reciprocal equalities are checked in `PrimePoleCriterion.lean`. Both are at source commit `25e960b69ea1346ffc511b199abf4cc755eb6df8`. The local exact scan supplies the complete maximal-hit payloads and resulting cancellations. Source paths and result identifiers are in `research_packet.json`. These local identities and the manuscript revision require the same public checkpoint.

Source-current fixed companion orbit. The complete infinite rationality boundary is checked in `CompanionOrbitRationality.lean`. Its paper-facing endpoints are `not_irrational_factorialGapSeries_iff_eventually_companion_floor_neg_two` and `irrational_factorialGapSeries_iff_cofinal_companion_floor_misses`; the generic fixed-orbit theorem, canonical factorial expansion, positive sub-unit exponential tail, and reverse telescoping assembly are in the same module. The focused source build and axiom print are green. This source-current module, the two-theorem Comparator revision, and this manuscript revision must be published at one common immutable checkpoint before external replay or Palomar submission readiness is claimed.

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